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Transforming User Stories to Requirements

Movie Shop Case Study

Learning Objectives

By completing this lab, you will be able to:

- 1 Identify and extract the three key components of a user story: Role (Actor), Feature (Function), and Benefit (Object)
- 2 Transform informal user stories into formal, SMART requirements
- 3 Apply the SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound) to requirements
- 4 Recognize how a single user story can expand into multiple detailed requirements

Part 1: Understanding the Movie Shop Domain

The Movie Shop is a retail business that sells and rents both physical and digital videos. The system must support multiple actors (customers, members, sales clerks, and managers) with various functionalities. Before transforming user stories, familiarize yourself with the key entities and actors in this domain:

Key Actors:

- Customer (Non-member): Can purchase/rent physical videos, view reviews, submit reviews
- Member: Pays annual fee, gets 10% discount, can make reservations, access digital videos
- Sales Clerk: Manages transactions, customer information, video inventory
- Shop Manager: Generates reports, oversees operations
- System: Handles video inventory, rental tracking, overdue notices

Key Entities:

- Physical Videos (copies with rental/sale tracking)
- Digital Videos (no physical copies)
- Customers/Members (with personal information)
- Rentals (with due dates and overdue tracking)
- Reservations (up to 5 per member)
- Reviews (with ratings 1-10, up to 100 words, optional anonymity)

Part 2: User Story Template and Structure

A well-formed user story follows this structure:

As a [ROLE/ACTOR] ,

I want to [FEATURE/FUNCTION] ,

so that [BENEFIT/OBJECT] .

Key Components:

Component	Definition	Example
Role (Actor)	Who is performing the action?	Customer, Member, Sales Clerk, Manager
Feature (Function)	What action or capability is needed?	Add to cart, make reservation, generate report
Benefit (Object)	Why is this needed? What value does it provide?	Purchase multiple items at once, ensure availability, track performance

Part 3: User Stories from Movie Shop Examples

Below are user stories extracted from the Movie Shop case study. Study each one and identify the Role, Feature, and Benefit.

User Story 1: Member Discount on Purchases

As a member,



I want to receive a 10% discount on video sales and rentals,
so that I can save money on my purchases and feel valued for my membership.

Role (Actor): Member

Feature (Function): Receive discount on sales and rentals

Benefit (Object): Save money, feel valued

User Story 2: Make Reservations Online

As a member,
I want to make reservations for physical video rentals via the Web,
so that I can secure videos I want without visiting the shop.

Role (Actor): Member

Feature (Function): Make reservations via Web

Benefit (Object): Secure videos in advance, convenience

User Story 3: Submit Video Reviews

As a customer,
I want to submit a review and rating for videos I have purchased or rented,
so that I can share my opinion and help other customers make purchasing decisions.

Role (Actor): Customer

Feature (Function): Submit review and rating

Benefit (Object): Share opinions, help other customers

User Story 4: Anonymous Reviews

As a customer,
I want to choose whether my name appears with my review,



so that I can protect my privacy while still sharing my feedback.

Role (Actor): Customer

Feature (Function): Choose whether their name appears with their review

Benefit (Object): Protect privacy while still sharing feedback

User Story 5: Track Overdue Rentals

As a shop manager,

I want the system to automatically track overdue physical video rentals,

so that I can ensure timely collection and maintain inventory control.

Role (Actor): Shop Manager

Feature (Function): Automatically track overdue physical video rentals

Benefit (Object): Ensure timely collection and maintain inventory control

User Story 6: Manage Customer Information

As a sales clerk,

I want to enter and update customer information (name, address, phone, age, sex, email),

so that I can maintain accurate customer records for sales and communication.

Role (Actor): Sales Clerk

Feature (Function): Enter and update customer information

Benefit (Object): Maintain accurate customer records for sales and communication

User Story 7: Look Up Customer Status

As a sales clerk,

I want to quickly look up whether a customer is a member,



so that I can apply appropriate discounts during transactions.

Role (Actor): Sales Clerk

Feature (Function): Look up customer membership status

Benefit (Object): Apply correct discounts, ensure accuracy

User Story 8: Manage Video Inventory

As a sales clerk,

I want to enter and update video information (title, director, genre, price, etc.),

so that I can maintain an accurate and searchable video catalog.

Role (Actor): Sales Clerk

Feature (Function): Enter and update video information

Benefit (Object): Maintain an accurate and searchable video catalog

User Story 9: Generate Sales Reports

As a shop manager,

I want to generate reports on video sales and rentals,

so that I can analyze business performance and make informed decisions.

Role (Actor): Shop Manager

Feature (Function): Generate reports on video sales and rentals

Benefit (Object): Analyze business performance and make informed decisions

User Story 10: Send Overdue Notices

As the system,

I want to automatically send email notices to customers with overdue physical video rentals,

so that customers are reminded to return videos and the shop can recover inventory.



Role (Actor): System

Feature (Function): Automatically send email notices for overdue rentals

Benefit (Object): Remind customers to return videos and help the shop recover inventory

Part 5: Exercises

Exercise 1: Extract Components from User Stories

For each user story below, identify and write down the Role (Actor), Feature (Function), and Benefit (Object):

User Story A:

As a customer,

I want to purchase digital videos online,

so that I can watch movies immediately without leaving home.

Your Answer:

- Role: Customer
 - Feature: Purchase digital videos online
 - Benefit: Watch movies immediately without leaving home
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User Story B:

As a sales clerk,

I want to process the return of rented physical videos,

so that I can update inventory and check for damages.

Your Answer:

- Role: Sales Clerk
 - Feature: Process the return of rented physical videos
 - Benefit: Update inventory and check for damages
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User Story C:

As a member,

I want to change my personal information online using my membership number and password,

so that I can keep my account up to date.

Your Answer:

- Role: Member
- Feature: Change personal information online using membership number and password
- Benefit: Keep account up to date

Part 4: Transformation Examples

Example 1: Member Discount Transformation

Original User Story:

As a member,

I want to receive a 10% discount on video sales and rentals,

so that I can save money on my purchases and feel valued for my membership.

Extracted Components:

- Role: Member
- Feature: Receive discount on sales and rentals
- Benefit: Save money, feel valued

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-1	The system must apply a 10% discount to all video sales for members.	S: Specific discount rate (10%); M: Discount amount calculated; A: Achievable with pricing logic; R: Supports business model; T: Applied at point of sale
	The system must apply a 10% discount to all video sales for members at point of sale	



Req ID	Requirement	SMART Criteria
REQ-2	The system must clearly display the discount amount during checkout.	S: Display discount; M: Discount shown in transaction; A: UI implementation; R: Transparency; T: At checkout
	The systems must display the discount during checkout in the transaction details page.	
REQ-3	The system must verify membership status before applying discounts.	S: Check membership; M: Status confirmed; A: Database lookup; R: Accuracy; T: Before discount applied
	The system must verify membership status by checking the membership database before applying the membership discount	

Example 2: Make Reservations Online Transformation

Original User Story:

As a member,
I want to make reservations for physical video rentals via the Web,
so that I can secure videos I want without visiting the shop.

Extracted Components:

- Role: Member
- Feature: Make reservations via Web
- Benefit: Secure videos in advance, convenience

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-5	Members must be able to reserve physical videos through the Web interface.	S: Web-based reservation; M: Reservation created; A: Web development; R: Convenience; T: Available 24/7
REQ-6	A member can reserve a maximum of 5 physical videos at any one time.	S: Maximum 5 reservations; M: Count enforced; A: Business rule; R: Inventory management; T: Per reservation attempt



Req ID	Requirement	SMART Criteria
REQ-7	The system must confirm reservations and display confirmation details to the member.	S: Confirmation provided; M: Confirmation displayed; A: UI implementation; R: User confirmation; T: Immediately after reservation
REQ-8	The system must notify members when reserved videos become available for pickup.	S: Notification sent; M: Email/notification delivered; A: Notification system; R: Customer engagement; T: When video available
REQ-9	Members must be able to make reservations by phone or in person in addition to the Web.	S: Multiple channels; M: Reservations accepted via phone/in-person; A: Staff training; R: Accessibility; T: During business hours

Example 3: Submit Video Reviews Transformation

Original User Story:

As a customer,
I want to submit a review and rating for videos I have purchased or rented,
so that I can share my opinion and help other customers make purchasing decisions.

Extracted Components:

- Role: Customer
- Feature: Submit review and rating
- Benefit: Share opinions, help other customers

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-10	The system must allow customers to submit reviews for videos they have purchased or rented.	S: Review submission; M: Review stored; A: Database storage; R: Customer engagement; T: After purchase/rental
REQ-11	Reviews must include a rating from 1 (lowest) to 10 (highest).	S: Rating scale 1-10; M: Rating value validated; A: Input validation; R: Standardized feedback; T: Required field



Req ID	Requirement	SMART Criteria
REQ-12	Reviews must be limited to a maximum of 100 words.	S: 100-word limit; M: Word count enforced; A: Text validation; R: Concise feedback; T: Before submission
REQ-13	The system must display reviews to other customers when browsing videos.	S: Display reviews; M: Reviews visible; A: UI implementation; R: Inform purchasing decisions; T: On video detail page
REQ-14	Customers must be able to choose whether their name appears with their review.	S: Anonymity option; M: Name display toggled; A: User preference; R: Privacy protection; T: At submission

Example 4: Track Overdue Rentals Transformation

Original User Story:

As a shop manager,
I want the system to automatically track overdue physical video rentals,
so that I can ensure timely collection and maintain inventory control.

Extracted Components:

- Role: Shop Manager
- Feature: Track overdue rentals automatically
- Benefit: Ensure timely collection, maintain inventory

Transformed into SMART Requirements:

Req ID	Requirement	SMART Criteria
REQ-15	The system must automatically identify physical video rentals that are past their due date.	S: Identify overdue rentals; M: Overdue status flagged; A: Date comparison logic; R: Inventory management; T: Daily check
REQ-16	The system must maintain a list of all overdue rentals accessible to the shop manager.	S: Overdue list available; M: List generated; A: Database query; R: Management visibility; T: Real-time
REQ-17	The system must send email notices to customers with overdue rentals.	S: Email notification; M: Email delivered; A: Email system; R: Customer reminder; T: When overdue



Req ID	Requirement	SMART Criteria
REQ-18	The system must allow the shop manager to generate overdue rental reports.	S: Report generation; M: Report created; A: Reporting module; R: Analysis and tracking; T: On demand
REQ-19	The system must track which copy of a physical video is rented and when it is due back.	S: Track copy and due date; M: Data recorded; A: Database storage; R: Inventory accuracy; T: At rental

Exercise 2: Transform User Stories to SMART Requirements

Transform the following user story into at least 5 SMART requirements. Use the format provided:

User Story:

As a member,
I want to track my rental history,
so that I can remember which videos I have rented and when.

Your Transformed Requirements:

Req ID	Requirement	SMART Criteria
REQ-1	The system must maintain a complete rental history for each member, recording the video title, rental date, and return date.	S: specific fields recorded; M: history entries countable/verifiable; A: standard database logging; R: supports member convenience; T: updated at time of each rental/return.
REQ-2	The system must allow members to view their rental history online at any time.	S: online access; M: list retrievable and displayed; A: standard web/app feature; R: member convenience; T: available 24/7 on demand.
REQ-3	Rental history must be sorted by most recent rental date by default.	S: specific sort order; M: order verifiable; A: standard sorting logic; R: usability; T: applied whenever history is displayed.
REQ-4	The system must display the due date and return status (returned, overdue, or current) for each entry in the rental history.	S: specific status values; M: status field checkable; A: derivable from existing due-date data; R: helps member avoid overdue fees; T: status shown in real time.
REQ-5	The system must retain a member's rental history for a minimum of 24 months.	S: specific retention duration; M: countable in months; A: standard data retention policy; R: supports long-term record keeping; T: continuous from each transaction date.



Exercise 3: Identify Missing Requirements

The following user story has been transformed into requirements, but some important requirements are missing. Identify what is missing and add 2-3 additional requirements:

Original User Story:

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As a sales clerk,
I want to sell physical videos to customers,
so that I can process transactions and maintain inventory.
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Existing Requirements:

- REQ-1: The system must allow sales clerks to select a video for sale.
- REQ-2: The system must calculate the sale price for the video.
- REQ-3: The system must process payment from the customer.

What is missing?

The existing requirements cover selecting a video, calculating price, and processing payment, but they omit applying the member discount, updating inventory after the sale, and giving the customer proof of purchase — all necessary to fully "process the transaction and maintain inventory" as the user story states.

Your Additional Requirements:

Req ID	Requirement
REQ-4	The system must apply the member discount (10%) automatically when the customer is a verified member, before finalizing the sale price.
REQ-5	The system must decrease the available inventory count for the sold physical video copy immediately after the sale is completed.
REQ-6	The system must generate and provide a receipt (printed or digital) confirming the transaction details to the customer.



Exercise 4: Create Your Own User Stories (Project Based)

Based on the Movie Shop case study, create new user stories for your project. For each story, ensure it follows the proper format and includes clear Role, Feature, and Benefit.

User Story 1:

As a social player,

I want to see other players' verified skill ratings before accepting a match,

so that I can choose a game at my level and feel confident about who I'm playing with.

User Story 2:

As a venue manager,

I want to set a temporary discount on a specific time slot from my dashboard,

so that I can fill empty off-peak courts without contacting each player manually.

User Story 3:

As a recurring group organizer,

I want to be notified if my team's regular booking slot becomes unavailable,

so that I can rebook a similar time before the group loses their weekly spot.

Part 4: Group Project

Exercise 4. 1: Generate all user stories of your project

All nine user stories generated for PlayGCC, grouped by persona (see Lab 2 personas: Sara the Social Player, Faisal the Recurring Group Organizer, and Munira the Venue Manager):

Sara Al-Otaibi — The Social Player

- US-1: As a social player, I want to find a verified player nearby within minutes, so that I don't have to cancel my game when my usual partner can't make it.

- US-2: As a female player, I want to filter for women-only sessions, so that I can play in a setting I feel comfortable in.
- US-7: As a social player, I want to see other players' verified skill ratings before accepting a match, so that I can choose a game at my level and feel confident about who I'm playing with.

Faisal Al-Harbi — The Recurring Group Organizer

- US-3: As a recurring group organizer, I want to set up an automatic weekly booking, so that I don't have to manually reserve and message everyone every week.
- US-4: As a group organizer, I want the cost split and auto-charged to each player, so that I'm not stuck fronting the bill.
- US-9: As a recurring group organizer, I want to be notified if my team's regular booking slot becomes unavailable, so that I can rebook a similar time before the group loses their weekly spot.

Munira Al-Qahtani — The Venue Manager

- US-5: As a venue manager, I want to see demand forecasts for my courts, so that I can set targeted discounts for slow time slots.
- US-6: As a venue manager, I want a real-time booking dashboard, so that my staff no longer manage reservations by phone or spreadsheet.
- US-8: As a venue manager, I want to set a temporary discount on a specific time slot from my dashboard, so that I can fill empty off-peak courts without contacting each player manually.

Exercise 4.2: Transform to requirements

Each user story from Exercise 4.1 was transformed into two SMART requirements, following the same method used in Part 4's worked examples.

Req ID	Requirement	SMART Criteria
REQ-1	The system must return at least one verified, available player match for a requested sport and time window within 2 minutes of a Quick Match request.	S: specific match action; M: response time measurable (2 min); A: feasible with existing player database query; R: solves the “short a player” problem; T: within 2 minutes of request.
REQ-2	The system must only suggest matches from players who have completed identity verification.	S: verification required; M: verified flag checked; A: uses existing Nafath-based verification; R: core trust requirement; T: checked at match-suggestion time.
REQ-3	The system must allow a player to filter Quick Match and open-session results to women-only sessions.	S: specific filter; M: filter toggle on/off verifiable; A: standard UI filter; R: serves the women's segment; T: applied at time of search.
REQ-4	The system must allow venues to tag specific time slots as women-only when configuring their schedule.	S: specific tag; M: tag stored per slot; A: venue dashboard feature; R: supply-side enabler for the filter; T: set at slot-creation time.

Req ID	Requirement	SMART Criteria
REQ-5	The system must allow a user to create a recurring booking for a fixed day and time that repeats weekly for up to 12 weeks.	S: specific recurrence rule; M: number of weeks countable; A: standard scheduling logic; R: solves repeat re-booking; T: repeats weekly until end date.
REQ-6	The system must automatically notify the organizer if a recurring booking fails to renew due to lack of availability.	S: specific failure case; M: notification sent/not sent verifiable; A: event-driven notification; R: keeps organizer informed; T: immediately when a renewal fails.
REQ-7	The system must divide the total booking cost evenly among all confirmed players by default, with the option for the organizer to adjust individual shares.	S: even-split default with override; M: amounts sum to total, verifiable; A: standard payment-split logic; R: removes the “one person fronts the bill” pain point; T: calculated at booking confirmation.
REQ-8	The system must send each player a payment request via the app or WhatsApp within 5 minutes of booking confirmation.	S: specific channel and timing; M: request sent/received logged; A: existing WhatsApp Business API integration; R: ensures timely collection; T: within 5 minutes.
REQ-9	The system must generate a weekly demand forecast per court showing predicted utilization by hour.	S: specific output (hourly, weekly, per court); M: forecast values verifiable against actual bookings; A: uses historical booking data; R: supports pricing decisions; T: refreshed weekly.
REQ-10	The system must flag time slots with a utilization forecast below 40% as “discount candidates” on the venue dashboard.	S: specific threshold (40%); M: flag present/absent; A: rule-based on forecast output; R: directs manager attention to low-utilization slots; T: updated with each forecast refresh.
REQ-11	The system must display real-time booking status for every court on the venue dashboard, updated within 5 seconds of any booking action.	S: real-time, specific latency; M: latency measurable; A: standard real-time sync; R: replaces the phone/spreadsheet process; T: within 5 seconds.
REQ-12	The system must allow multiple staff accounts per venue with role-based permissions (view-only vs. edit).	S: specific roles defined; M: permission enforcement testable; A: standard access control; R: supports multi-staff front-desk use; T: enforced on every dashboard action.
REQ-13	The system must display a player's skill rating, derived from match history and partner ratings, on their profile before a match is accepted.	S: specific data shown; M: rating value present/verifiable; A: uses existing ratings data; R: builds confidence before playing a stranger; T: visible at match-review time.
REQ-14	The system must recalculate a player's skill rating after every completed and rated match.	S: specific trigger; M: rating value change verifiable; A: standard aggregation logic; R: keeps ratings accurate over time; T: immediately after each rated match.
REQ-15	The system must allow a venue manager to apply a percentage discount to a specific court and time slot directly from the dashboard.	S: specific action (percentage, slot); M: discount value stored/verifiable; A: standard dashboard feature; R: fills off-peak slots without manual outreach; T: applied immediately upon save.
REQ-16	The system must automatically surface discounted slots to nearby players as a “deal” within the booking view.	S: specific surfacing behavior; M: deal visible/not visible; A: existing search/filter logic; R: connects the supply-side discount to demand; T: shown as soon as the discount is active.
REQ-17	The system must notify a recurring booking's organizer at least 24 hours before a renewal if the original slot is no longer available.	S: specific timing (24 hrs); M: notification sent/not sent; A: checked against availability before renewal; R: gives the organizer time to react; T: 24 hours prior to the affected booking.



Req ID	Requirement	SMART Criteria
REQ-18	The system must suggest up to 3 alternative time slots at the same venue when a recurring booking cannot be renewed.	S: specific number of alternatives; M: suggestions count verifiable; A: uses existing availability search; R: helps the organizer keep the group's weekly habit; T: shown at the time of the renewal-failure notification.

Exercise 4.3: Review the requirements to be SMART.

All 18 requirements above satisfy the SMART criteria:

- **Specific:** each requirement names one concrete system behavior (e.g. “return a verified match,” “flag slots below 40% utilization”) rather than a vague goal.
- **Measurable:** each requirement includes a value that can be checked — a time limit, a count, a threshold, or a present/absent flag.
- **Achievable:** every requirement relies on functionality already scoped in the market study and business model canvas (matchmaking data, Nafath verification, WhatsApp API, dashboard), not speculative technology.
- **Relevant:** each requirement traces directly back to a persona's stated goal or pain point from Lab 2, so nothing was added that doesn't serve a real user need.
- **Time-bound:** each requirement specifies when or how often the behavior occurs (e.g. “within 2 minutes,” “weekly,” “24 hours prior”).

Transforming the user stories also surfaced requirements that were not obvious from the stories alone — for example, US-4 (“split and auto-charge the cost”) only became testable once it was broken into a specific split rule (REQ-7) and a specific notification channel and timing (REQ-8). This matches the “Completeness” takeaway in Part 6: the transformation process itself revealed missing functionality that the original informal user stories had left implicit.

Part 6: Key Takeaways

Why Transform User Stories to Requirements?

- 5 Clarity: User stories are informal and conversational; requirements are formal and precise
- 6 Implementation Guidance: Requirements provide specific details for developers
- 7 Testing: Requirements can be tested; user stories are too vague for testing
- 8 Traceability: Requirements can be tracked and verified
- 9 Completeness: Transformation often reveals missing functionality



SMART Criteria Explained

Criterion	Definition	Example
Specific	Clear and well-defined, not ambiguous	"Apply 10% discount" not "Apply a discount"
Measurable	Can be quantified or verified	"Maximum 5 reservations" (countable)
Achievable	Realistic and feasible within constraints	Consider technical and business feasibility
Relevant	Aligned with business goals and user needs	Supports the Movie Shop's business model
Time-bound	Specifies when or how often	"At point of sale," "Within 1 second," "Daily"

Common Mistakes to Avoid

- 10 Too Vague: "The system should be user-friendly" → Too subjective
- 11 Too Technical: Jumping to implementation details instead of requirements
- 12 Missing Context: Not considering who, what, and why
- 13 Incomplete: Forgetting edge cases and error handling
- 14 Unmeasurable: Requirements that cannot be tested or verified

Submissions :

- 1- Lab 3 doc
- 2- Project Features, including Lab 2 Personas and user stories.